

A. Orac and LCM

time limit per test: 3 seconds
memory limit per test: 256 megabytes

For the multiset of positive integers $s = \{s_1, s_2, \dots, s_k\}$, define the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of s as follow:

- $\gcd(s)$ is the maximum positive integer x , such that all integers in s are divisible on x .
- $\text{lcm}(s)$ is the minimum positive integer x , that divisible on all integers from s .

For example, $\gcd(\{8, 12\}) = 4$, $\gcd(\{12, 18, 6\}) = 6$ and $\text{lcm}(\{4, 6\}) = 12$. Note that for any positive integer x , $\gcd(\{x\}) = \text{lcm}(\{x\}) = x$.

Orac has a sequence a with length n . He come up with the multiset $t = \{\text{lcm}(\{a_i, a_j\}) \mid i < j\}$, and asked you to find the value of $\gcd(t)$ for him. In other words, you need to calculate the GCD of LCMs of all pairs of elements in the given sequence.

Input

The first line contains one integer n ($2 \leq n \leq 100\,000$).

The second line contains n integers, $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 200\,000$).

Output

Print one integer: $\gcd(\{\text{lcm}(\{a_i, a_j\}) \mid i < j\})$.

Examples

input 2 1 1	output 1
input 4 10 24 40 80	output 40
input 10 540 648 810 648 720 540 594 864 972 648	output 54

Note

For the first example, $t = \{\text{lcm}(\{1, 1\})\} = \{1\}$, so $\gcd(t) = 1$.

For the second example, $t = \{120, 40, 80, 120, 240, 80\}$, and it's not hard to see that $\gcd(t) = 40$.